

实验名称 实验8 BA无标度网络的生成及拓扑特征分析

一、实验内容及结果

- 1、利用 BA 无标度网络生成算法分别生成 N 相同， m (m_0) 不同的 10 个网络，计算每个网络的平均度、平均路径长度、聚类系数、余平均度、余度分布，分析随着 m (m_0) 的增加，这些拓扑参数的变化趋势。
- 2、利用 BA 无标度网络生成算法分别生成 m (m_0) 相同， N 不同的 10 个网络，计算每个网络的平均度、平均路径长度、聚类系数、余平均度、余度分布，分析随着 N 的增加，这些拓扑参数的变化趋势。

源代码：

[ba 无标度网络生成算法]

```
import networkx as nx
import random
import matplotlib.pyplot as plt
def barabasi_albert_graph(n0,n,m):
    G=nx.complete_graph(n0)
    targets=list(range(m)) #0-m1
    repeated_nodes=list(range(n0))*m
    source=n0

    while source<n:
        G.add_edges_from(zip([source]*m,targets))
        repeated_nodes.extend(targets)
        repeated_nodes.extend([source]*m)
        targets=random.sample(repeated_nodes,m)
        targets=set(targets)
        while len(targets)<m:
            x= random.choice(repeated_nodes)
            targets.add(x)
        source+=1
    return G
```

[余平均度函数]

```
def average_nearest_neighbor_degree(G):
    k=set([G.degree(i) for i in G.nodes()])
    sorted_k=sorted(k)

    k_nn_k=[]
    for ki in sorted_k:
        if ki != 0:
            c=0
            k_nn_i=0
            for i in G.nodes():
                if G.degree(i)==ki:
                    k_nn_i+=sum([G.degree(j) for j in list(nx.all_neighbors(G,i))])/ki
            c+=1
            k_nn_k.append(k_nn_i/c)

    m=sorted(k)
    p=k_nn_k
    edd = {}
    S=0
    for i in range(len(m)):
        S+=p[i]
    a_knn=S/len(m)

    return a_knn
```

[余度分布函数]

```
def neighbor_distribution(G):
    all_k=[G.degree(i,weight='weight') for i in G.nodes()]
    k=list(set(all_k))
    N=len(G.nodes())
    d=dict(nx.degree(G))
    avd=sum(d.values())/len(G.nodes)

    Pk=[]
    Qk=[]
    for ki in sorted(k):
        c=0
        for i in G.nodes():
            if G.degree(i,weight='weight')==ki:
                c+=1
        pk=c/N
```

```

        Pk.append(pk)
        qk=ki*pk/avd
        Qk.append(qk)

    return sorted(k),Pk,Qk

```

[生成 10 个 m 不同的网络]

```

G=[]
N0=15
N=100
random.seed(1)
M=random.sample(range(N0),10)
M=sorted(M)
print(M)
networks = []
for m in M:
    G=barabasi_albert_graph(N0,N,m)
    networks.append(G)
    nx.draw(G,node_color="green",edge_color="red",with_labels=True,node_s
size=50)
    plt.title("BA scale-free networkx",fontsize=20)
    plt.show()

```

[N 不变 m 变化的拓扑参数]

```

AVD={}
L={}
AVT={}
AE={}
random.seed(1)
M=random.sample(range(N0),10)
M=sorted(M)

for i, G in enumerate(networks):
    d=dict(nx.degree(G))
    #平均度
    avd=sum(d.values())/len(G.nodes)
    AVD[i]=avd
    print(f"Graph {i+1}:")
    print(f"M={M[i]}:")
    # print(M)
    print(f"平均度: {avd}")

    #最短路径
    if nx.is_connected(G):

```

```

        l=nx.average_shortest_path_length(G)
    else:
        l=0
    L[i]=l
    print(f"最短路径: {l}")

    #聚类系数
    avt=nx.average_clustering(G)
    AVT[i]=avt
    print(f"聚类系数: {avt}")

    #余平均度
    a_knn= average_nearest_neighbor_degree(G)
    AE[i]=a_knn
    print(f"余平均度:{a_knn}")
    print(" ")

```

[余度分布和绘图]

```

for i, G in enumerate(networks):
    k,P_dis,A_dis=neighbor_distribution(G)
    print(f"M={M[i]}:")
    print(f"余度分布: ")
    for ki, ad in zip(k,A_dis):
        print(f" Degree {ki}: {ad}")
    print("")
    print(f"余度分布的和为{sum(A_dis)} ")

    degrees = k
    A_values = A_dis

    # 绘制关系图
    plt.figure(figsize=(10, 6))
    plt.bar(degrees, A_values, alpha=0.5,color='green')
    plt.xlabel('Degree (k)')
    plt.ylabel('A value')
    plt.title(f'Excess Degree Distribution for M={M[i]}')
    plt.grid(True)
    plt.show

```

[各拓扑参数的变化趋势]

```

plt.plot(M, AVD.values(), marker='o', linestyle='-')
plt.xlabel('M')
plt.ylabel('AVD')
plt.title('<k> vs M')

```

```

plt.grid(True)
plt.show()

plt.plot(M, L.values(), marker='o', linestyle='--')
plt.xlabel('M')
plt.ylabel('L')
plt.title('L vs M')
plt.grid(True)
plt.show()

plt.plot(M, AVT.values(), marker='o', linestyle='--')
plt.xlabel('M')
plt.ylabel('AVT')
plt.title('C vs M')
plt.grid(True)
plt.show()

plt.plot(M, AE.values(), marker='o', linestyle='--')
plt.xlabel('M')
plt.ylabel('AE')
plt.title('AE vs M')
plt.grid(True)
plt.show()

import matplotlib.pyplot as plt
degrees = []
A_values = []

for i, G in enumerate(networks):
    k, D_dis, A_dis = neighbor_distribution(G)
    degrees.append(k)
    A_values.append(A_dis)

plt.figure(figsize=(10, 6))

for i in range(len(M)):
    plt.bar(degrees[i], A_values[i], label=f"m={M[i]}")

plt.xlabel('Degree (k)')
plt.ylabel('A value')
plt.title('Excess Degree Distribution for each m')
plt.legend()
plt.grid(True)
plt.show()

```

[生成 10 个 N 不同的网络]

```
Gp=[]
N0=15
M=10
random.seed(10)
N=random.sample(list(range(15, 150, 10)), 10)
N=sorted(N)
print(N)
networksp = []
for n in N:
    Gp=barabasi_albert_graph(N0,n,M)
    networksp.append(Gp)
    nx.draw(Gp,node_color="green",edge_color="red",with_labels=True,node_
size=50)
    plt.title("BA scale-free networkx",fontsize=20)
    plt.show()
```

[m 相同 N 不同的拓扑参数]

```
random.seed(10)
N=random.sample(list(range(15, 150, 10)), 10)
N=sorted(N)
AVD={}
L={}
AVT={}
AE={}

for i, G in enumerate(networksp):
    d=dict(nx.degree(G))
    #平均度
    avd=sum(d.values())/len(G.nodes)
    AVD[i]=avd
    print(f"Graph {i+1}:")
    print(f"N= {N[i]}")
    print(f"平均度: {avd}")

    #最短路径
    if nx.is_connected(G):
        l=nx.average_shortest_path_length(G)
    else:
        l=0
    L[i]=l
    print(f"最短路径: {l}")

    #聚类系数
```

```

avt=nx.average_clustering(G)
AVT[i]=avt
print(f"聚类系数: {avt}")

#余平均度
a_knn= average_nearest_neighbor_degree(G)
AE[i]=a_knn
print(f"余平均度:{a_knn}")
print(" ")

```

[余度分布及绘图]

```

print(f"余度分布: ")
for i, G in enumerate(networksp):
    k,D_dis,A_dis=neighbor_distribution(G)
    print(f"N= {N[i]}:")
    for ki, ad in zip(k,A_dis):
        print(f" Degree {ki}: {ad}")
    print("")
    print(f"余度分布的和为{sum(A_dis)} ")

degrees = k
A_values = A_dis

# 绘制关系图
plt.figure(figsize=(10, 6))
plt.bar(degrees, A_values, alpha=0.5,color='blue')
plt.xlabel('Degree (k)')
plt.ylabel('A value')
plt.title(f'Excess Degree Distribution for N={N[i]}')
plt.grid(True)
plt.show()

```

[各拓扑参数的变化趋势]

```

plt.plot(N, AVD.values(), marker='o', linestyle='--')
plt.xlabel('N')
plt.ylabel('AVD')
plt.title('<k> vs N')
plt.grid(True)
plt.show()

plt.plot(N, L.values(), marker='o', linestyle='--')
plt.xlabel('N')
plt.ylabel('L')

```

```

plt.title('L vs N')
plt.grid(True)
plt.show()

plt.plot(N, AVT.values(), marker='o', linestyle='--')
plt.xlabel('N')
plt.ylabel('AVT')
plt.title('C vs N')
plt.grid(True)
plt.show()

plt.plot(N, AE.values(), marker='o', linestyle='--')
plt.xlabel('N')
plt.ylabel('AE')
plt.title('AE vs N')
plt.grid(True)
plt.show()

import matplotlib.pyplot as plt

degrees = []
A_values = []

for i, G in enumerate(networksp):
    k,D_dis,A_dis = neighbor_distribution(G)
    degrees.append(k)
    A_values.append(A_dis)

plt.figure(figsize=(10, 6))

for i in range(len(N)):
    plt.bar(degrees[i], A_values[i], label=f"N={N[i]}")

plt.xlabel('Degree (k)')
plt.ylabel('A value')
plt.title('Excess Degree Distribution for each N')
plt.legend()
plt.grid(True)
plt.show()

```

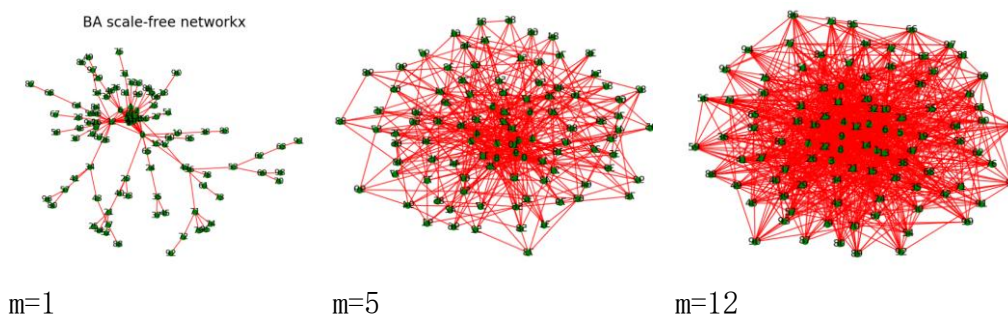
运行结果 (截图):

TASK1:

N 相同, m 不同时的各网络的平均度、平均路径长度、聚类系数、余平均度:

令 $N_0=15$, $N=100$, 随机生成的 M 列表为 [1, 2, 3, 4, 5, 7, 8, 9, 11, 12]。
初始 N_0 设置大于 10, 因为要 10 个网络, m 大于 10, 要保证初始时新加入节点至少连接 m 条边而不产生重边, 则初始节点数 N_0 应大于 10。

【网络形态】



【拓扑参数】

Graph 1:
M=1:
平均度: 3.8
最短路径: 3.974141414141414
聚类系数: 0.1176947311088178
余平均度: 10.217632077197349

Graph 2:
M=2:
平均度: 5.5
最短路径: 2.874343434343434
聚类系数: 0.2285820596567399
余平均度: 10.935604574187536

Graph 3:
M=3:
平均度: 7.2
最短路径: 2.534949494949495
聚类系数: 0.2566634643378857
余平均度: 13.041520297504443

Graph 4:
M=4:
平均度: 8.9
最短路径: 2.334343434343434
聚类系数: 0.21535045492353258
余平均度: 14.421314798680385

Graph 5:
M=5:
平均度: 10.6
最短路径: 2.184040404040404
聚类系数: 0.23037313301666276
余平均度: 16.2684141087552

Graph 6:
M=7:
平均度: 14.0
最短路径: 2.013939393939394
聚类系数: 0.26903120288329285
余平均度: 19.010711997947784

Graph 7:
M=8:
平均度: 15.7
最短路径: 1.9418181818181819
聚类系数: 0.28322808957656453
余平均度: 21.142486398160194

Graph 8:
M=9:
平均度: 17.4
最短路径: 1.8937373737373737
聚类系数: 0.3003123070841404
余平均度: 22.878059324123388

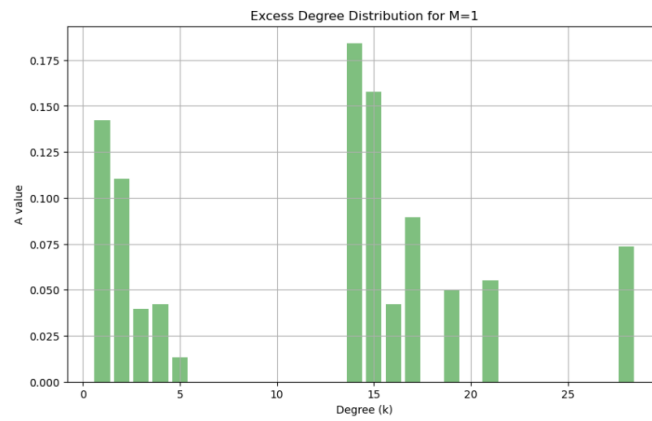
Graph 9:
M=11:
平均度: 20.8
最短路径: 1.8149494949494949
聚类系数: 0.32629791701976674
余平均度: 26.284331755110177

Graph 10:
M=12:
平均度: 22.5
最短路径: 1.7836363636363637
聚类系数: 0.3528011241453309
余平均度: 28.11712246139563

N 相同，m 不同时的余度分布：

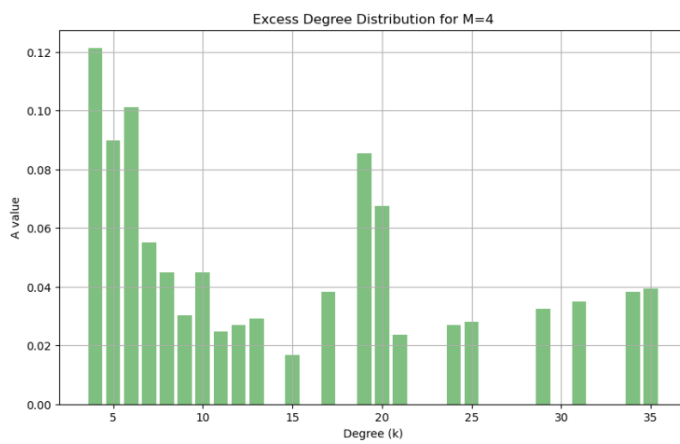
M=1:
余度分布：
Degree 1: 0.14210526315789476
Degree 2: 0.11052631578947368
Degree 3: 0.03947368421052632
Degree 4: 0.04210526315789474
Degree 5: 0.013157894736842106
Degree 14: 0.1842105263157895
Degree 15: 0.15789473684210525
Degree 16: 0.04210526315789474
Degree 17: 0.08947368421052633
Degree 19: 0.05
Degree 21: 0.05526315789473684
Degree 28: 0.0736842105263158

余度分布的和为1.0000000000000002



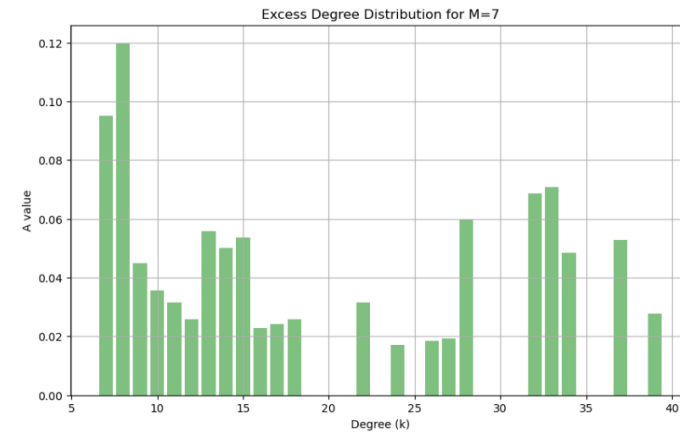
M=4:
余度分布：
Degree 4: 0.12134831460674157
Degree 5: 0.0898876404494382
Degree 6: 0.10112359550561796
Degree 7: 0.0550561797752809
Degree 8: 0.0449438202247191
Degree 9: 0.030337078651685393
Degree 10: 0.0449438202247191
Degree 11: 0.024719101123595506
Degree 12: 0.026966292134831458
Degree 13: 0.029213483146067417
Degree 15: 0.016853932584269662
Degree 17: 0.038202247191011236
Degree 19: 0.0853932584269663
Degree 20: 0.06741573033707865
Degree 21: 0.023595505617977526
Degree 24: 0.026966292134831458
Degree 25: 0.02808987640449437
Degree 29: 0.03258426966292134
Degree 31: 0.0348314606741573
Degree 34: 0.038202247191011236
Degree 35: 0.03932584269662921

余度分布的和为1.0000000000000002



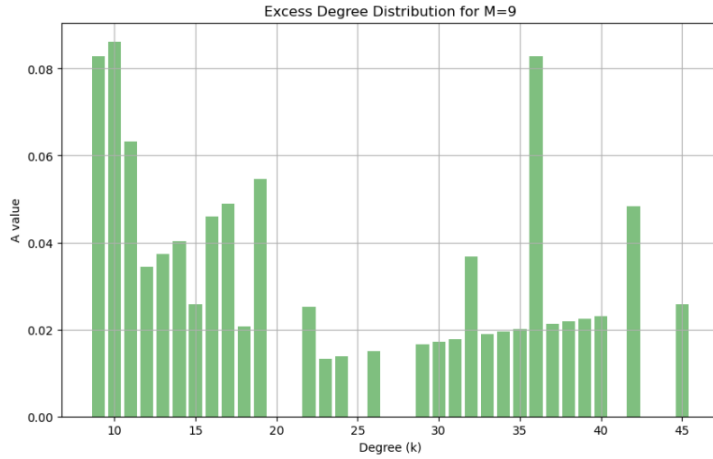
M=7:
余度分布：
Degree 7: 0.095
Degree 8: 0.12
Degree 9: 0.045000000000000005
Degree 10: 0.03571428571428571
Degree 11: 0.03142857142857143
Degree 12: 0.025714285714285714
Degree 13: 0.055714285714285716
Degree 14: 0.05
Degree 15: 0.05357142857142857
Degree 16: 0.022857142857142857
Degree 17: 0.02428571428571429
Degree 18: 0.025714285714285714
Degree 22: 0.03142857142857143
Degree 24: 0.017142857142857144
Degree 26: 0.018571428571428572
Degree 27: 0.019285714285714288
Degree 28: 0.06
Degree 32: 0.06857142857142857
Degree 33: 0.07071428571428572
Degree 34: 0.04857142857142858
Degree 37: 0.05285714285714286
Degree 39: 0.027857142857142858

余度分布的和为1.0



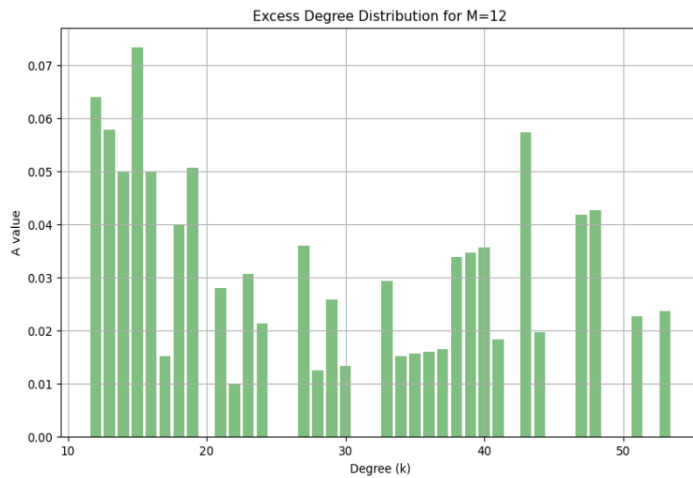
M=9:
 余度分布:
 Degree 9: 0.08275862068965517
 Degree 10: 0.08620689655172414
 Degree 11: 0.06321839080459771
 Degree 12: 0.03448275862068966
 Degree 13: 0.03735632183908046
 Degree 14: 0.04022988505747127
 Degree 15: 0.02586206896551724
 Degree 16: 0.04597701149425288
 Degree 17: 0.04885057471264369
 Degree 18: 0.020689655172413793
 Degree 19: 0.0545977011494253
 Degree 22: 0.025287356321839084
 Degree 23: 0.013218390804597703
 Degree 24: 0.013793103448275862
 Degree 26: 0.014942528735632185
 Degree 29: 0.016666666666666666
 Degree 30: 0.017241379310344827
 Degree 31: 0.017816091954022988
 Degree 32: 0.036781609195402305
 Degree 33: 0.018965517241379314
 Degree 34: 0.019540229885057474
 Degree 35: 0.020114942528735635
 Degree 36: 0.08275862068965517
 Degree 37: 0.021264367816091957
 Degree 38: 0.021839080459770118
 Degree 39: 0.02241379310344828
 Degree 40: 0.02298850574712644
 Degree 42: 0.04827586206896552
 Degree 45: 0.025862068965517244

余度分布的和为1.0



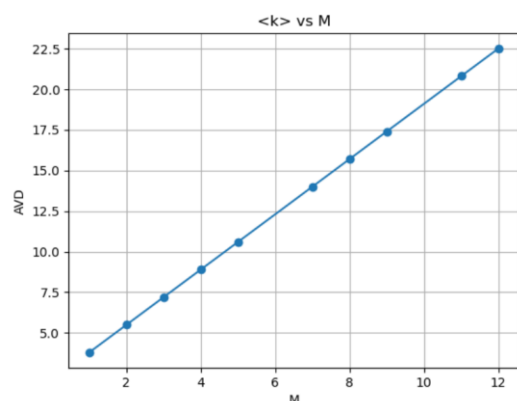
M=12:
 余度分布:
 Degree 12: 0.064
 Degree 13: 0.05777777777777778
 Degree 14: 0.04977777777777778
 Degree 15: 0.07333333333333333
 Degree 16: 0.04977777777777778
 Degree 17: 0.015111111111111112
 Degree 18: 0.04
 Degree 19: 0.05066666666666665
 Degree 21: 0.028
 Degree 22: 0.009777777777777778
 Degree 23: 0.03066666666666665
 Degree 24: 0.021333333333333333
 Degree 27: 0.036
 Degree 28: 0.012444444444444445
 Degree 29: 0.025777777777777774
 Degree 30: 0.013333333333333332
 Degree 33: 0.029333333333333336
 Degree 34: 0.015111111111111112
 Degree 35: 0.015555555555555557
 Degree 36: 0.016
 Degree 37: 0.016444444444444446
 Degree 38: 0.03377777777777778
 Degree 39: 0.034666666666666665
 Degree 40: 0.035555555555555556
 Degree 41: 0.018222222222222223
 Degree 43: 0.057333333333333333
 Degree 44: 0.019555555555555555
 Degree 47: 0.04177777777777778
 Degree 48: 0.042666666666666665
 Degree 51: 0.022666666666666667
 Degree 53: 0.023555555555555555

余度分布的和为1.0

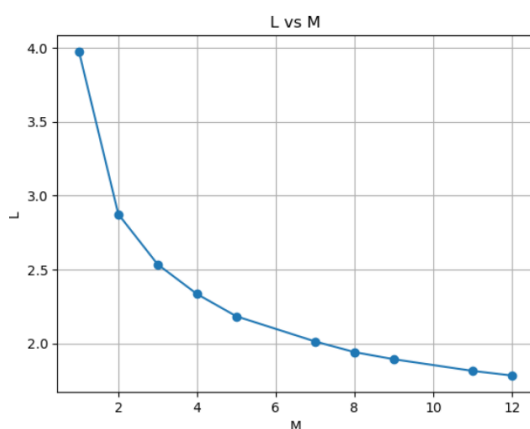


从 10 个网络中选取了五个，左图是余度分布具体数值，右图是不同连变数 M 下的余度分布图。计算可知，对每个网络，各个度下余度分布的和都为 1。

基本拓扑性质随 m 变化趋势 ($M=[1, 2, 3, 4, 5, 7, 8, 9, 11, 12]$):

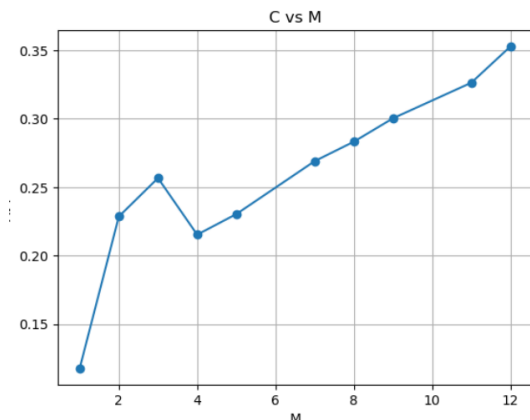


左图是平均度随 m 增加的变化趋势，可以看出平均度随 m 的增加而增加，两者呈近似线性正相关关系。



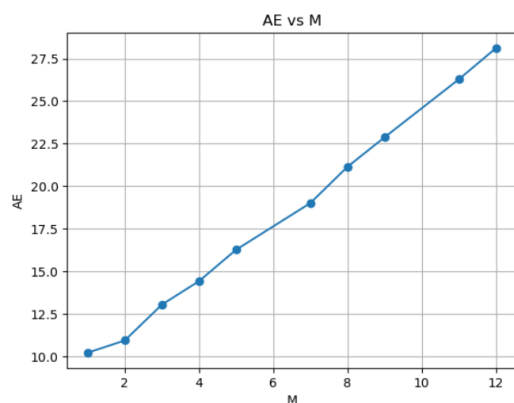
左图是平均路径长度随 m 增加的变化趋势。平均路径长度整体上随着 m 的增大而减小，且减小的速率由快到慢。

因为随着每次新加入节点形成连边数的增加，网络中的短路径更容易快速形成，后加入节点的新边对最短路径影响小。



左图是聚类系数随 m 增加的变化趋势。整体而言，聚类系数随 m 的增加而增加。

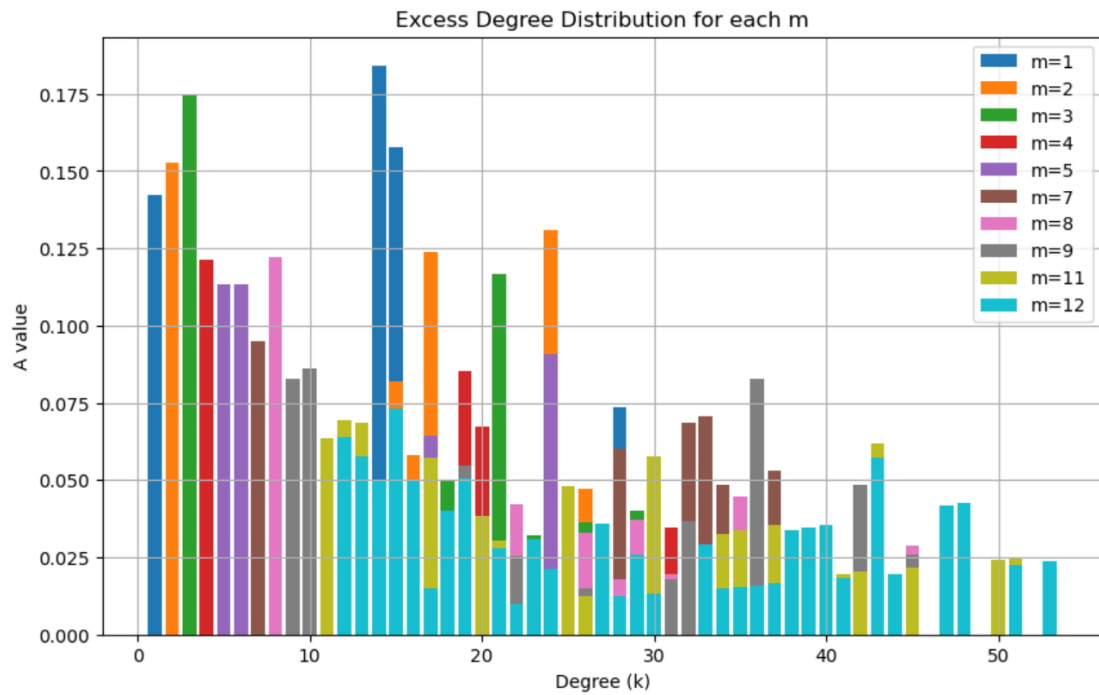
因为聚类系数衡量的是节点的邻居互为邻居的概率，由于 ba 无标度网络的优先连接特性，每次连接的边数增多，连接到同时连接某节点和其邻居的可能性也增大。



左图是余平均度随 m 增加的变化趋势。余平均度随 m 的增加而增大。

网络余平均度反映网络中平均每个节点的邻居节点的平均度。由于 ba 无标度网络的优先连接特性，新加入节点倾向和度大的节点连接，则计算余平均度时度大的节点会作为很多点的邻居被多次计算。

余度分布随 M 变化趋势：



这是不同 M 下节点的余度分布，观察图像，可以获知，随着 m 的增加，**余度分布涉及的节点的度范围变大**，纵向对比前文单个 m 下的余度分布图也可知，随 m 增加**分布变密集**。因为 m 增加，边连接的随机性也增加，最终得到节点的度数值更多样。

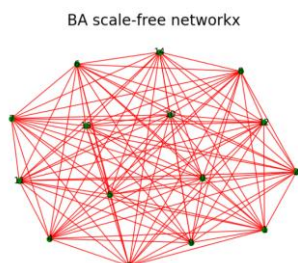
TASK2:

m 相同, N 不同时的各网络的平均度、平均路径长度、聚类系数、余平均度:

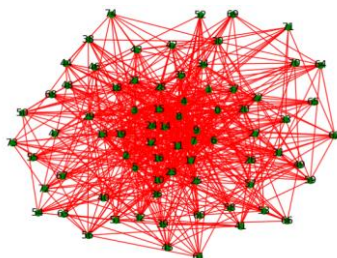
令 $N_0=15, M=10$, 随机生成的最大网络规模 N 为:

[15, 35, 45, 75, 85, 105, 115, 125, 135, 145]

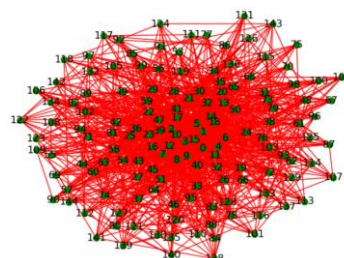
【网络形态】



N=15



N=75



N=145

【拓扑参数】

Graph 1:

N= 15

平均度: 14.0

最短路径: 1.0

聚类系数: 1.0

余平均度: 14.0

Graph 6:

N= 105

平均度: 19.142857142857142

最短路径: 1.8573260073260074

聚类系数: 0.302358770404749

余平均度: 24.65582897967084

Graph 2:

N= 35

平均度: 17.428571428571427

最短路径: 1.4890756302521009

聚类系数: 0.6302426213501728

余平均度: 19.43768717652258

Graph 7:

N= 115

平均度: 19.217391304347824

最短路径: 1.8889397406559878

聚类系数: 0.28422125523344693

余平均度: 25.256528713582494

Graph 3:

N= 45

平均度: 18.0

最短路径: 1.5929292929292929

聚类系数: 0.5189508089006386

余平均度: 20.413716478293228

Graph 8:

N= 125

平均度: 19.28

最短路径: 1.9054193548387097

聚类系数: 0.3011604987417597

余平均度: 26.07994279997954

Graph 4:

N= 75

平均度: 18.8

最短路径: 1.76

聚类系数: 0.38236188413275424

余平均度: 23.175117859993083

Graph 9:

N= 135

平均度: 19.333333333333332

最短路径: 1.9354339414040906

聚类系数: 0.2470645718727129

余平均度: 25.478512514525736

Graph 5:

N= 85

平均度: 18.941176470588236

最短路径: 1.796358543417367

聚类系数: 0.3548132727220249

余平均度: 23.781881084805953

Graph 10:

N= 145

平均度: 19.379310344827587

最短路径: 1.9586206896551723

聚类系数: 0.24122465705087354

余平均度: 26.479696410559246

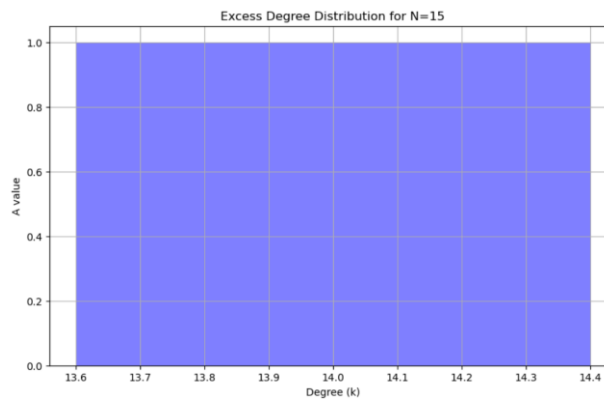
N 相同，m 不同时的余度分布：

余度分布：

N= 15:

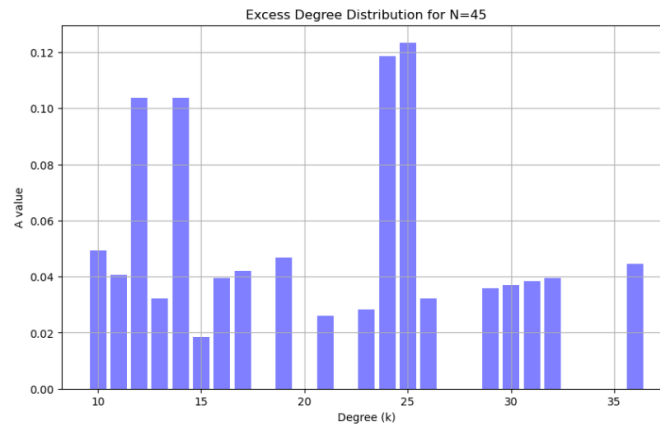
Degree 14: 1.0

余度分布的和为1.0



N= 45:

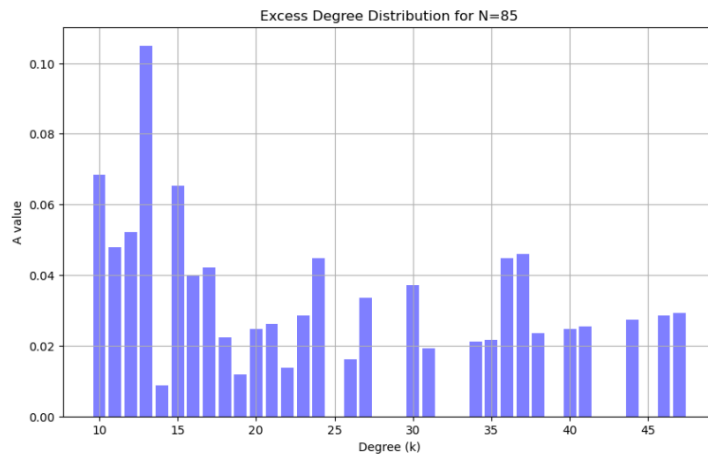
Degree 10: 0.04938271604938272
Degree 11: 0.04074074074074074
Degree 12: 0.1037037037037037
Degree 13: 0.03209876543209877
Degree 14: 0.1037037037037037
Degree 15: 0.01851851851851852
Degree 16: 0.03950617283950617
Degree 17: 0.04197530864197531
Degree 19: 0.04691358024691358
Degree 21: 0.025925925925925925
Degree 23: 0.028395061728395066
Degree 24: 0.11851851851851852
Degree 25: 0.1234567901234568
Degree 26: 0.03209876543209877
Degree 29: 0.03580246913580247
Degree 30: 0.03703703703703704
Degree 31: 0.03827160493827161
Degree 32: 0.03950617283950617
Degree 36: 0.04444444444444444



余度分布的和为0.9999999999999999

N= 85:

Degree 10: 0.06832298136645963
Degree 11: 0.04782608695652174
Degree 12: 0.052173913043478265
Degree 13: 0.10496894409937889
Degree 14: 0.008695652173913044
Degree 15: 0.06521739130434782
Degree 16: 0.03975155279503106
Degree 17: 0.0422360248447205
Degree 18: 0.022360248447204967
Degree 19: 0.011801242236024844
Degree 20: 0.024844720496894408
Degree 21: 0.026086956521739132
Degree 22: 0.013664596273291923
Degree 23: 0.02857142857142857
Degree 24: 0.044720496894409933
Degree 26: 0.016149068322981366
Degree 27: 0.03354037267080745
Degree 30: 0.03726708074534161
Degree 31: 0.019254658385093167
Degree 34: 0.02111801242236025
Degree 35: 0.021739130434782608
Degree 36: 0.044720496894409933
Degree 37: 0.04596273291925466
Degree 38: 0.023602484472049687
Degree 40: 0.024844720496894408
Degree 41: 0.02546583850931677
Degree 44: 0.027329192546583846
Degree 46: 0.02857142857142857
Degree 47: 0.029192546583850933

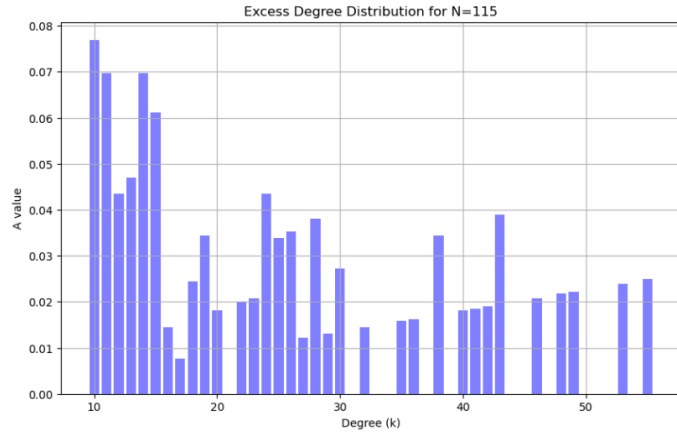


余度分布的和为1.0

N= 115:

Degree 10: 0.07692307692307694
Degree 11: 0.06968325791855204
Degree 12: 0.04343891402714933
Degree 13: 0.04705882352941177
Degree 14: 0.06968325791855204
Degree 15: 0.061085972850678745
Degree 16: 0.014479638009049774
Degree 17: 0.007692307692307694
Degree 18: 0.024434389140271493
Degree 19: 0.034389140271493215
Degree 20: 0.01809954751131222
Degree 22: 0.01990950226244344
Degree 23: 0.020814479638009052
Degree 24: 0.04343891402714933
Degree 25: 0.03393665158371041
Degree 26: 0.03529411764705882
Degree 27: 0.012217194570135748
Degree 28: 0.03800904977375566
Degree 29: 0.013122171945701358
Degree 30: 0.027149321266968326
Degree 32: 0.014479638009049774
Degree 35: 0.015837104072398193
Degree 36: 0.016289592760180997
Degree 38: 0.034389140271493215
Degree 40: 0.01809954751131222
Degree 41: 0.018552036199095023
Degree 42: 0.01900452488687783
Degree 43: 0.038914027149321274
Degree 46: 0.020814479638009052
Degree 48: 0.021719457013574663
Degree 49: 0.022171945701357467
Degree 53: 0.02398190045248869
Degree 55: 0.0248868778280543

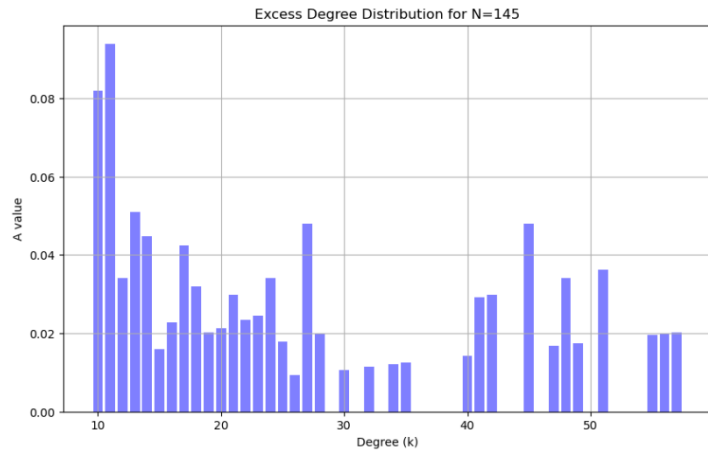
余度分布的和为1.000000000000002



N= 145:

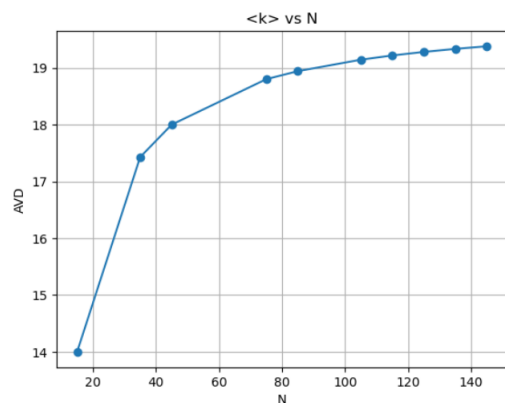
Degree 10: 0.08185053380782918
Degree 11: 0.09395017793594305
Degree 12: 0.034163701067615654
Degree 13: 0.050889679715302484
Degree 14: 0.04483985765124555
Degree 15: 0.01601423487544484
Degree 16: 0.02277580071174377
Degree 17: 0.04234875444839857
Degree 18: 0.03202846975088968
Degree 19: 0.020284697508896797
Degree 20: 0.021352313167259784
Degree 21: 0.0298932384341637
Degree 22: 0.023487544483985764
Degree 23: 0.024555160142348754
Degree 24: 0.034163701067615654
Degree 25: 0.017793594306049824
Degree 26: 0.009252669039145907
Degree 27: 0.04804270462633452
Degree 28: 0.0199288256227758
Degree 30: 0.010676156583629892
Degree 32: 0.011387900355871885
Degree 34: 0.012099644128113879
Degree 35: 0.012455516014234874
Degree 40: 0.014234875444839857
Degree 41: 0.029181494661921707
Degree 42: 0.0298932384341637
Degree 45: 0.04804270462633452
Degree 47: 0.01672597864768683
Degree 48: 0.034163701067615654
Degree 49: 0.017437722419928824
Degree 51: 0.036298932384341634
Degree 55: 0.019572953736654804
Degree 56: 0.0199288256227758
Degree 57: 0.020284697508896797

余度分布的和为1.0

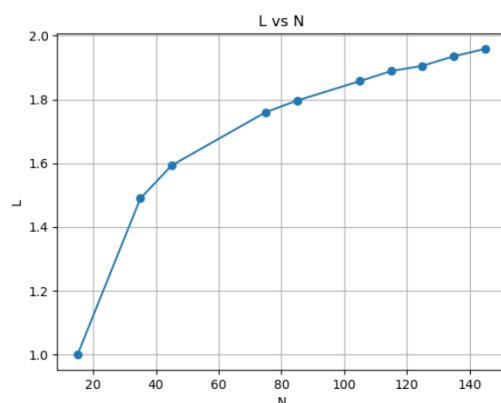


从 10 个网络中选取了五个，左图是余度分布具体数值，右图是不同最大网络节点数 N 下的余度分布图。计算可知，对每个网络，各个度下余度分布的和都为 1。

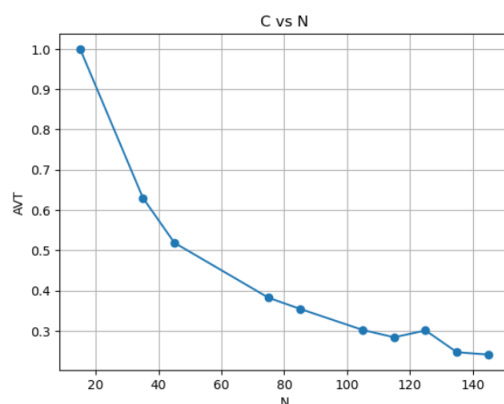
基本拓扑性质随 N 变化趋势 ($N = [15, 35, 45, 75, 85, 105, 115, 125, 135, 145]$) :



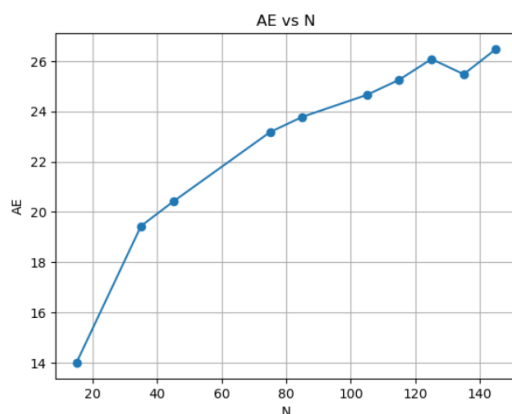
左图是平均度随 N 增加的变化趋势，可以看出**平均度随 N 增加而增加，且增长速率变缓**。因为随着网络规模的增加，每个节点的连接数逐渐趋于饱和，平均度增速下降。



左图是平均路径长度随 N 增加的变化趋势，**平均路径长度随着 N 的增大而增大，增速也有变缓趋势**。因为随着网络最大节点数增加，网络中任意两节点间的路径长度也大概率增加，从而平均路径长增加。

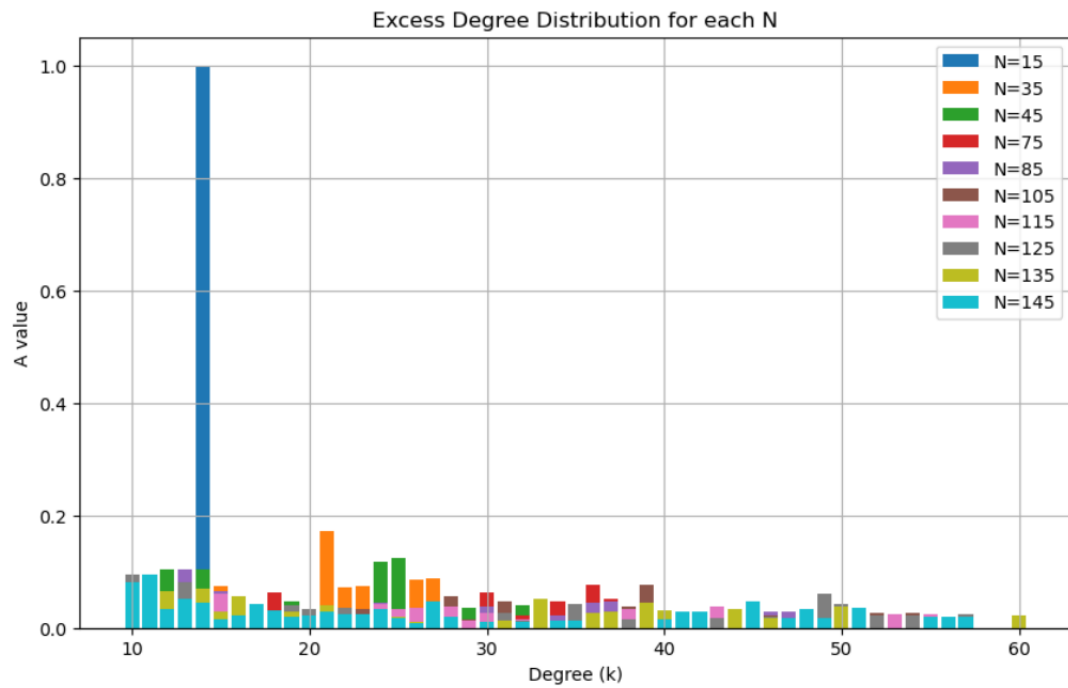


左图是聚类系数随 N 增加的变化趋势。总体上**聚类系数随 N 的增加而减小**。因为聚类系数衡量的是节点的邻居互为邻居的概率每次连接的边数不变，但网络规模增大，节点的邻居互为邻居的概率减小。



左图是余平均度随 N 增加的变化趋势。**余平均度随 N 的增加而增大**。因为网络余平均度反映网络中平均每个节点的邻居节点的平均度，网络规模增大，ba 无标度网络具有优先连接特性，新加入节点倾向和度大的节点连接，则计算余平均度时度大的节点会作为很多点的邻居被多次计算。

余度分布随 M 变化趋势：



随着网络规模 N 的增加，网络的余度分布**范围更广，分布更分散**。当 $M=15$ 时，余度分布在度为 14 的节点上的概率为 1，而当 $M=145$ 时，节点的邻居的度广泛分布在 10-57 区间范围内。